GGG-W-646
AMENDMENT-3
JANUARY 15, 1962

SUPERSEDING
Amendment-2
May 10, 1960

FEDERAL SPECIFICATION

WRENCH, OPEN END, RATCHET (TAC PATTERN), FOR TUBE FITTINGS, ELECTRIC CABLE TERMINALS AND STUFFING TUBE GLAND NUTS

This amendment, which forms a part of Federal Specification GGG-W-646, dated July 13, 1954, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 1, paragraph 1.2, line 9: After "Open end" add "(electronic cable fitting wrench)".

Page 1, paragraph 2.1: Add the following:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Military Specification:

MIL-P-15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Page 2, paragraph 2.1: Under *Military Standards* add:

MIL-STD-129—Marking for Shipment and Storage.

Page 2, paragraph 2.1: Delete parenthetical paragraphs and substitute the following:

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Page 2, paragraph 2.2: Delete in its entirety.

Page 2, paragraph 3.2.1, line 3, after "cracks," add "blowholes,".

Page 2, paragraph 3.2.2, sentences 2 and 3: Delete and substitute "Unless otherwise specified (see 6.1), wrenches shall have one or a combination of the coatings listed herein;

FSC 5120

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if chromium and nickel-plate coatings are specified but are restricted by Government order, one of the other coatings listed will be acceptable. The rotating member or ratchet of the wrench head and the hexagon drive of the internal wrenching attachments shall be phosphate or oxide coated."

Page 5, table II, column 5, item 10: Delete ".995" and substitute ".955".

Page 6, table II, column 5, item 11: Delete "2.470" and substitute "2.473".

Page 7, table III: Delete and substitute new table III.

Page 8, paragraph 3.5.1.2.2, line 1: After "*open end*" add "(*electronic cable fitting wrench*)".

Page 8, figure 3, title: After "open end" add "(electronic cable fitting wrench)".

Page 8, paragraph 3.6.1: Delete the last two sentences and substitute: "The sockets shall be made of steel and shall be free of all pits, nodules, burrs, cracks, blowholes, and other detrimental defects. The nut engaging surfaces shall be smooth. The sockets shall have a Rockwell hardness of 40 to 52 on the "C" scale."

Page 10, table V, column 7, item 20: Delete "1.825" and substitute "1.875".

Page 12, table VI, columns 5, 6, 8, and 10: Delete and substitute:

AA		CC ± .032	EE ± .032
Minimum	Maximum		
Inches	*Inches*	*Inches*	*Inches*
1.056	1.066	1.437	1.109
1.150	1.160	1.562	1.109
1.275	1.285	1.687	1.109
1.400	1.411	1.812	1.250
1.468	1.474	1.875	1.250
1.681	1.693	2.187	1.265
1.806	1.818	2.312	1.265
1.900	1.910	2.437	1.281
1.962	1.975	2.500	1.281
2.213	2.228	2.812	1.281
2.275	2.290	2.875	1.359
2.369	2.385	3.000	1.359
2.494	2.510	3.125	1.359
2.838	2.856	3.500	1.375
2.994	3.013	3.625	1.390
3.119	3.139	3.750	1.453
3.306	3.326	4.000	1.453
3.463	3.484	4.187	1.468
3.587	3.609	4.312	1.468
3.775	3.798	4.500	1.468

Page 12, table VI, footnote: Delete.

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TABLE III.—*Type I, class 1, style B, form B, open ratchet, crowfoot single-purpose; open-end (electronic cable fitting wrench) wrench heads*

Component number	Rated torque	Nominal wrench size	GG[1]	Dimensions						
				NN		JJ[1]	LL +0.015 −.005	MM +0.005 −.000	KK	
				Minimum	Maximum				Degrees	Minutes
	Inch-pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*		
33	150	3/8	1.380	0.378	0.386	1.820	0.322	0.254	7	30
34	180	7/16	1.380	.440	.448	1.820	.322	.254	7	30
35	200	1/2	1.380	.504	.512	1.820	.322	.254	7	30
36	225	9/16	1.630	.566	.575	2.200	.352	.381	7	30
37	380	5/8	1.933	.629	.638	2.433	.362	.381	7	30
38	420	11/16	2.053	.692	.701	2.593	.372	.381	7	30
39	460	3/4	2.173	.755	.765	2.733	.400	.381	7	30
40	520	13/16	2.301	.818	.828	2.901	.410	.381	7	30
41	675	7/8	2.421	.880	.890	3.021	.420	.381	7	30
42	730	15/16	2.561	.944	.955	3.161	.425	.381	7	30
43	840	1	2.685	1.006	1.017	3.285	.435	.381	7	30
194	930	1 1/16	2.807	1.068	1.080	3.407	.460	.381	7	30
195	930	1 1/8	2.807	1.132	1.145	3.407	.460	.381	7	30
196	970	1 3/16	2.987	1.194	1.207	3.687	.490	.508	7	30
44	970	1 1/4	2.987	1.257	1.270	3.687	.490	.508	7	30
197	1030	1 5/16	3.107	1.320	1.334	3.807	.490	.508	7	30
45	1030	1 3/8	3.107	1.383	1.397	3.807	.490	.508	7	30
198	1100	1 7/16	3.307	1.446	1.460	4.007	.490	.508	7	30
46	1100	1 1/2	3.307	1.508	1.523	4.007	.490	.508	7	30
199	1175	1 9/16	3.467	1.571	1.586	4.167	.490	.508	7	30
47	1175	1 5/8	3.467	1.634	1.649	4.167	.490	.508	7	30
200	1250	1 11/16	3.627	1.696	1.711	4.327	.490	.508	6	0
48	1250	1 3/4	3.627	1.759	1.775	4.327	.490	.508	6	0
201	1300	1 13/16	3.692	1.822	1.838	4.392	.490	.508	6	0
49	1300	1 7/8	3.692	1.885	1.901	4.392	.490	.508	6	0
202	1350	1 15/16	3.807	1.948	1.964	4.527	.490	.508	6	0
50	1350	2	3.807	2.011	2.028	4.527	.490	.508	6	0
203	1400	2 1/16	3.957	2.074	2.092	4.707	.570	.508	6	0
51	1400	2 1/8	3.957	2.136	2.155	4.707	.570	.508	6	0
204	1500	2 3/16	4.117	2.200	2.219	4.867	.580	.508	5	0
52	1500	2 1/4	4.117	2.262	2.281	4.867	.580	.508	5	0
205	1500	2 5/16	4.440	2.324	2.344	5.190	.580	.508	5	0
53	1500	2 3/8	4.440	2.388	2.408	5.190	.580	.508	5	0
206	1500	2 7/16	4.607	2.453	2.491	5.367	.600	.508	5	0
54	1500	2 1/2	4.607	2.513	2.534	5.367	.600	.508	5	0
207	1500	2 9/16	4.780	2.576	2.587	5.540	.680	.508	5	0
55	1500	2 5/8	4.780	2.639	2.660	5.540	.690	.508	5	0
208	1500	2 11/16	4.957	2.701	2.722	5.737	.750	.508	5	0
56	1500	2 3/4	4.957	2.766	2.787	5.737	.750	.508	5	0
209	1500	2 13/16	4.957	2.627	2.849	5.737	.750	.508	5	0
57	1500	2 7/8	4.957	2.890	2.912	5.737	.750	.508	5	0
210	1500	2 15/16	4.957	2.954	2.977	5.737	.750	.508	5	0
58	1500	3	4.957	3.016	3.039	5.737	.750	.508	5	0

[1]Nominal dimension, held within Drop-forging Association tolerances.

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Page 13, table VIII, column 8, heading: Delete "9" and substitute "8".

Page 16, under *"Nomenclature"* column, line 2: Delete "standard" and substitute "thin".

Pages 19 and 20, paragraphs 4.2 through 4.4.1: Delete "Government" wherever it appears.

Page 20: Add as paragraph 4.5:

4.5 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Pages 20 and 21: Delete section 5 and substitute new section 5:

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 *Level A.*—Level A preservation and packaging shall be in accordance with level A of Military Specification MIL-P-15424.

5.1.2 *Level C.*—Level C preservation and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Level A packing shall be in accordance with level A packing of Military Sepecification MIL-P-15424.

5.2.2 *Level B.*—Level B packing shall be in accordance with level B packing of Military Specification MIL-P-15424.

5.2.3 *Level C.*—Level C wrenches preserved and packaged in accordance with level A or C as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. This level in general shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements.

5.3 Marking.

5.3.1 *Civil agencies.* — Unless otherwise specified (see 6.1), marking for shipment shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL-STD-129.

Page 21, paragraph 6.1, item (*f*): Delete and substitute:

(*f*) Whether preservation and packaging should be level A or C and whether packing should be level, A, B, or C (see section 5).

Page 21, paragraph 6.1: Add as item (*g*):

(*g*) Marking, if different (see 5.3.2).

Pages 21 and 22, section 7: Delete:

MILITARY CUSTODIANS:

Army—Q
Navy—ShMC
Air Force—MOA

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